



1. Pascal's Triangle

output:

```
      1
     1 1
    1 2 1
   1 3 3 1
  1 4 6 4 1
```

Code:

```
public class PascalsTriangle {
    public static void main(String[] args) {
        int rows = 5;
        for (int i = 0; i < rows; i++) {
            int number = 1;
            System.out.printf("%" + (rows - i) * 2 + "s", "");
            for (int j = 0; j <= i; j++) {
                System.out.printf("%4d", number);
                number = number * (i - j) / (j + 1);
            }
            System.out.println();
        }
    }
}
```



2. Hollow Pyramid Pattern

output:

```
      *
    *  *
  *    *
*      *
*****
```

Code:

```
public class HollowPyramid {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            for (int j = i; j < n; j++) System.out.print(" ");
            for (int k = 1; k <= (2 * i - 1); k++) {
                if (k == 1 || k == (2 * i - 1) || i == n)
                    System.out.print("*");
                else
                    System.out.print(" ");
            }
            System.out.println();
        }
    }
}
```



3. Number Pyramid Pattern

output:

```
    1
   2 3
  4 5 6
 7 8 9 10
11 12 13 14 15
```

Code:

```
public class NumberPyramid {
    public static void main(String[] args) {
        int n = 5, num = 1;
        for (int i = 1; i <= n; i++) {
            for (int j = i; j < n; j++) System.out.print(" ");
            for (int k = 1; k <= i; k++) {
                System.out.print(num++ + " ");
            }
            System.out.println();
        }
    }
}
```

4. Binary Number Triangle

output:

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

Code:

```
public class BinaryTriangle {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            int num = (i % 2 == 0) ? 0 : 1;
            for (int j = 1; j <= i; j++) {
                System.out.print(num + " ");
                num = 1 - num;
            }
            System.out.println();
        }
    }
}
```

5. Inverted Number Pyramid

output:

```
1 2 3 4 5
 2 3 4 5
   3 4 5
    4 5
     5
```

Code:

```
public class InvertedNumberPyramid {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            for (int s = 1; s < i; s++) System.out.print(" ");
            for (int j = i; j <= n; j++) System.out.print(j + " ");
            System.out.println();
        }
    }
}
```